ML with Python (part3)

K-fold Cross Validation

Using the train/test method of estimating the skill of the procedure on unseen data sometimes has a high variance. This means that when it is repeated, it gives different results, often very different results.

Cross-validation is another method to estimate the skill of a method on unseen data. Like using a train-test split. Cross-validation is a statistical method used to estimate the skill of machine learning models. It is commonly used in applied machine learning to compare and select a model for a given predictive modelling problem because it is easy to understand, and easy to implement.

This, in turn, provides a population of performance measures.

- We can calculate the mean of these measures to get an idea of how well the procedure performs on average.
- We can calculate the standard deviation of these measures to get an idea of how much the skill of the procedure is expected to vary in practice.

The train-test split and k-fold cross validation are called resampling methods. Resampling methods are statistical procedures for sampling a dataset and estimating an unknown quantity.

Your model will likely perform better when trained on all of the available data than just the subset used to estimate the performance of the model.

The general procedure is as follows:

1. Shuffle the dataset randomly.
2. Split the dataset into k groups
3. For each unique group:
   - Take the group as a hold out or test data set
   - Take the remaining groups as a training data set
   - Fit a model on the training set and evaluate it on the test set
   - Retain the evaluation score and discard the model
4. Summarize the skill of the model using the sample of model evaluation scores